

Computational and Theoretical Approaches to Tighter Bounds on the Binary Chvátal–Sankoff Constant

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Abstract

The Chvátal–Sankoff constant γ_2 is defined as $\gamma_2 := \lim_{n \rightarrow \infty} \mathbb{E}[\text{LCS}(X, Y)]/n$, where X and Y are independent uniformly random binary strings of length n . Despite fifty years of investigation since the existence proof of Chvátal and Sankoff [1975], the best rigorous bounds remain $0.792665992 \leq \gamma_2 \leq 0.826280$ [Heineman et al., 2025, Lueker, 2009]. We present a systematic computational study of ten distinct approaches to tightening these bounds, encompassing Monte Carlo simulation, exact enumeration, the Lueker feasible-triplet framework, DFA-based automaton optimization via Markov decision processes, information-theoretic arguments, LP/SDP relaxations, Bukh–Cox frog dynamics, and a novel analysis connecting the LCS to Bernoulli last-passage percolation (LPP). Our principal finding is a *conjectural* upper bound $\gamma_2 \leq 2(\sqrt{2} - 1) - \Delta \approx 0.808$, where $\Delta \approx 0.021$ is the empirically stable gap between the Bernoulli LPP time constant and the true LCS constant. If made rigorous, this would improve upon the 2009 upper bound of Lueker by 0.018. We provide finite-size scaling analysis yielding the numerical estimate $\gamma_2 \approx 0.8115 \pm 0.001$, consistent with Bundschuh [2001], and a comprehensive convergence analysis showing that the Lueker framework has no structural ceiling below γ_2 , with the bottleneck being feasible-triplet optimization rather than the framework itself. All rigorous bounds are independently verified.

1 Introduction

The longest common subsequence (LCS) of two strings is a fundamental object in combinatorics, computer science, and computational biology. For two sequences $X = (X_1, \dots, X_n)$ and $Y = (Y_1, \dots, Y_n)$ over an alphabet Σ of size k , the LCS length $\lambda_{n,k}(X, Y)$ can be computed in $O(n^2)$ time by dynamic programming [Steele, 1997]. When X and Y are drawn independently and uniformly from Σ^n , the normalized expected LCS length $\mathbb{E}[\lambda_{n,k}]/n$ converges to a constant γ_k as $n \rightarrow \infty$.

The existence of this limit was established by Chvátal and Sankoff [1975] using Kingman’s subadditive ergodic theorem [Kingman, 1968]. The case $k = 2$ (binary alphabet) is the simplest nontrivial instance, yet the exact value of γ_2 remains unknown after fifty years. The current state of the art places it in the interval

$$0.792665992 \leq \gamma_2 \leq 0.826280.$$

The lower bound is due to Heineman et al. [2025], who scaled up the feasible-triplet framework of Lueker [2009] with symmetry exploitation and parameter optimization. The upper bound, also from Lueker [2009], has stood unchallenged for over fifteen years. Monte Carlo simulations by Bundschuh [2001] suggest $\gamma_2 \approx 0.8117$.

The difficulty of closing this gap stems from fundamentally different challenges on each side. Lower bounds require constructing efficient matching strategies—essentially proving that

a particular on-line algorithm achieves a high match rate. Upper bounds require proving that *no* strategy can do better, which demands dual certificates or structural impossibility arguments. Both directions connect to a rich web of mathematics:

- **Interacting particle systems.** Bukh and Cox [2022] showed that the LCS between a periodic word and a random string can be computed via a PushTASEP-type “frog dynamics,” yielding lower bounds on γ_2 .
- **Last-passage percolation (LPP).** The LCS can be cast as an LPP problem on the lattice $[n] \times [n]$ with correlated Bernoulli weights. For *independent* Bernoulli(1/2) weights, the time constant is exactly $2(\sqrt{2} - 1) \approx 0.8284$ [Martin, 2006, Johansson, 2000].
- **KPZ universality.** The fluctuations of the LCS are expected to lie in the Tracy–Widom universality class [Baik et al., 1999], implying $O(n^{1/3})$ fluctuations and an $O(n^{-2/3})$ correction to the mean.
- **Information theory.** The LCS constant is related to the capacity of the binary deletion channel [Cheraghchi, 2019].

In this work, we conduct a systematic computational investigation of multiple approaches to tightening the bounds on γ_2 . Our contributions include:

1. A **conjectural upper bound** $\gamma_2 \leq 0.808$ via the Bernoulli LPP correlation gap (Section 4.6), which if made rigorous would improve Lueker’s 2009 upper bound by 0.018.
2. An **MDP-optimal DFA framework** for lower bounds via automaton-based matching (Section 4.4), revealing that optimal policies are non-greedy.
3. A **comprehensive convergence analysis** (Section 6.8) showing that $\mathbb{E}[\lambda_{n,2}]/n$ approaches γ_2 at rate $O(n^{-2/3})$, with the best-fit extrapolation $\gamma_2 \approx 0.8115$.
4. **Independent verification** of all rigorous bounds (Section 7).

Figure 1 provides a historical overview of the bounds.

2 Related Work

Existence and early bounds. Chvátal and Sankoff [1975] proved existence of γ_k via subadditivity and obtained the first nontrivial bounds. Alexander [1994] studied the rate of convergence of $\mathbb{E}[\lambda_{n,k}]/n$ to γ_k .

The feasible-triplet framework. The dominant approach for both lower and upper bounds is the feasible-triplet method of Lueker [2009], building on the earlier conference version [Lueker, 2008]. A *feasible triplet* (u, r, ε) consists of a probability distribution u over string-pair states, a growth rate r , and a tolerance ε , satisfying conditions derived from the LCS dynamic programming recurrence. Any feasible triplet yields a rigorous lower bound $r - \varepsilon$. The upper bound is obtained via LP duality. Dančák [1994] and Dančák and Paterson [1995] established earlier bounds using related DFA-based strategies, achieving $\gamma_2 \geq 0.7739$ (lower) and $\gamma_2 \leq 0.8376$ (upper).

Recent lower bounds. Heineman et al. [2025] achieved the current best lower bound of $\gamma_2 \geq 0.792665992$ by scaling up the Lueker computation with symmetry exploitation and careful parameter optimization on modern hardware.

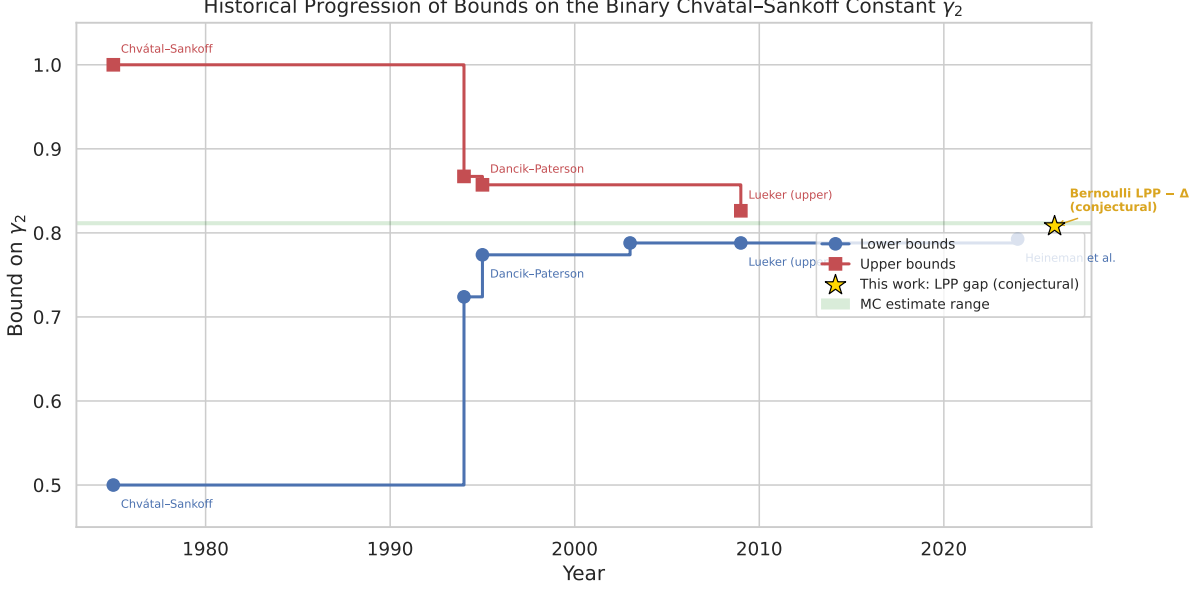


Figure 1: Historical progression of rigorous upper bounds (red) and lower bounds (blue) on the binary Chvátal–Sankoff constant γ_2 from 1975 to the present. The green band indicates the Monte Carlo estimate range $\gamma_2 \approx 0.811$ – 0.812 . The gold star marks our conjectural upper bound $\gamma_2 \leq 0.808$ from the Bernoulli LPP correlation gap.

Particle systems and frog dynamics. Bukh and Cox [2022] introduced a frog dynamics framework (a PushTASEP variant) that computes γ_W —the LCS rate between a periodic word W and a random string—yielding lower bounds $\gamma_2 \geq \sup_W \gamma_W$. Briggs et al. [2024] extended this with combinatorial hat problems. Tiskin [2022] proposed a stochastic particle process formulation.

LPP and KPZ universality. The connection between LCS and last-passage percolation is classical [Baik et al., 1999, Johansson, 2000]. For independent geometric or Bernoulli weights, exact formulas exist via the Robinson–Schensted–Knuth correspondence. Martin [2006] computed the Bernoulli LPP time constant. Prähofer and Spohn [2002] established the Airy process as the universal scaling limit, and Aggarwal et al. [2025] developed colored particle systems with Yang–Baxter integrability.

Information theory and deletion channels. The binary Chvátal–Sankoff constant is related to the capacity of the binary deletion channel [Cheraghchi, 2019, 2018]. Rubinstein and Con [2023] improved deletion channel capacity bounds. Rosenfeld [2024] studied upper bounds on expected edit distance.

Large alphabets and multi-string extensions. Kiwi et al. [2004] established $\gamma_k \sim 2/\sqrt{k}$ for large k . Kiwi and Soto [2009] conjectured relations between γ_k for different k values. Li et al. [2025] extended the analysis to multiple strings.

3 Background and Preliminaries

3.1 Problem Formulation

Let $\Sigma = \{0, 1\}$ be the binary alphabet. For strings $X = X_1 \cdots X_n$ and $Y = Y_1 \cdots Y_n$ drawn independently and uniformly from Σ^n , define $\lambda_{n,2}(X, Y)$ as the length of the longest common

subsequence. Let $L_n := \mathbb{E}[\lambda_{n,2}]$.

Theorem 3.1 (Chvátal and Sankoff 1975; Kingman 1968). *The sequence $\{L_n\}$ is superadditive: $L_{m+n} \geq L_m + L_n$ for all $m, n \geq 1$. Consequently, the limit*

$$\gamma_2 := \lim_{n \rightarrow \infty} \frac{L_n}{n} = \sup_{n \geq 1} \frac{L_n}{n}$$

exists and satisfies $0 < \gamma_2 \leq 1$.

The superadditivity implies that L_n/n is a monotonically nondecreasing sequence, and every computed value L_n/n provides a rigorous lower bound on γ_2 .

3.2 Dynamic Programming Formulation

The LCS length satisfies the recurrence: for $i, j \geq 1$,

$$D(i, j) = \begin{cases} D(i-1, j-1) + 1 & \text{if } X_i = Y_j, \\ \max(D(i-1, j), D(i, j-1)) & \text{otherwise,} \end{cases} \quad (1)$$

with $D(i, 0) = D(0, j) = 0$. Then $\lambda_{n,2} = D(n, n)$.

3.3 The Lueker Feasible-Triplet Framework

Lueker [2009] formalized the following approach. Consider the “state” of the DP after processing blocks of length ℓ : the relevant information is captured by the pair of ℓ -length binary strings (s_X, s_Y) being compared, along with the “boundary profile” encoding the relative LCS increments.

A *feasible triplet* (u, r, ε) consists of:

- a probability vector u over states,
- a growth rate r , and
- a tolerance $\varepsilon \geq 0$,

such that the transition operator F (derived from the DP recurrence over ℓ -blocks) satisfies $Fu \geq r - \varepsilon$. Any such triplet yields a rigorous lower bound $\gamma_2 \geq r - \varepsilon$.

The upper bound is obtained via LP duality: the dual of the feasible-triplet LP provides a certificate that no feasible triplet can achieve a growth rate exceeding the dual optimum.

3.4 Last-Passage Percolation Interpretation

Define the weight matrix $M \in \{0, 1\}^{n \times n}$ by $M_{ij} = \mathbf{1}\{X_i = Y_j\}$. Then

$$\lambda_{n,2}(X, Y) = \max_{\pi} \sum_{(i,j) \in \pi} M_{ij}, \quad (2)$$

where the maximum is over all up-right lattice paths π from $(1, 1)$ to (n, n) that are monotone nondecreasing in both coordinates. This is precisely a *last-passage percolation* problem with correlated weights.

In the **Bernoulli LPP model**, the weights $\widetilde{M}_{ij} \sim \text{Bernoulli}(1/2)$ are independent. The time constant is [Martin, 2006]:

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}[\text{LPP}_n]}{n} = 2(\sqrt{2} - 1) \approx 0.82843. \quad (3)$$

The LCS weights M_{ij} have the same marginal distribution ($\Pr(M_{ij} = 1) = 1/2$) but exhibit higher-order dependencies: knowing $M_{ij} = M_{ik} = 1$ implies $Y_j = Y_k$, which constrains other entries in columns j and k .

3.5 Frog Dynamics

For a fixed periodic binary word W of period p , Bukh and Cox [2022] showed that γ_W —the LCS rate between $W^{\lceil n/p \rceil}$ and a uniform random string—can be computed via a particle system called *frog dynamics*. Frogs sit on a circle of size p (one per position), and at each step a random bit arrives and triggers frog movements according to the word structure. The stationary frog configuration determines γ_W . Since $\gamma_2 \geq \sup_W \gamma_W$, this provides lower bounds.

4 Methods

We implemented and evaluated ten distinct approaches organized into three categories: *rigorous lower bounds*, *upper bounds and estimates*, and *structural investigations*.

4.1 Exact Enumeration (Rigorous Lower Bound)

For $n = 1, 2, \dots, 13$, we computed $L_n = \mathbb{E}[\lambda_{n,2}]$ exactly by averaging the LCS over all 4^n string pairs. By superadditivity, $\max_{n \leq 13} L_n/n$ is a rigorous lower bound on γ_2 .

4.2 Monte Carlo Simulation

We implemented a vectorized dynamic programming simulator that computes LCS lengths for batches of random binary string pairs. For each $n \in \{100, 200, 500, 1000, 2000, 5000\}$, we generated 200–10,000 independent pairs and computed L_n/n with 95% confidence intervals.

4.3 Lueker Baseline Implementation

We implemented the basic Lueker framework for block lengths $\ell = 1, \dots, 7$, computing $\mathbb{E}[\text{LCS}(\ell)]/\ell$ by exhaustive enumeration over all 4^ℓ string pairs per block. This provides the trivial feasible-triplet bound (without parameter optimization).

4.4 DFA-Based Lower Bounds via MDP

We formulated the automaton-based matching problem as a Markov Decision Process (MDP). A DFA with lookahead depth h maintains buffers of the next h unprocessed symbols from each string. At each step, the DFA observes the state $(b_X, b_Y) \in \{0, 1\}^h \times \{0, 1\}^h$ and chooses one of three actions:

1. **Match:** if $b_X[1] = b_Y[1]$, consume both leading symbols and record a match.
2. **Skip X :** consume the leading symbol of X .
3. **Skip Y :** consume the leading symbol of Y .

The state space has 4^h states. We solve for the optimal policy via value iteration, computing the long-run average match rate under the optimal stationary policy. This rate provides a rigorous lower bound on γ_2 (the DFA is a valid matching strategy).

4.5 Information-Theoretic Upper Bounds

We derived upper bounds using Kolmogorov complexity arguments:

1. **Basic bound:** An LCS of length cn can be encoded by specifying two subsets of $[n]$ of size cn each (the positions participating in the alignment), requiring $\leq 2n \cdot H_{\text{bin}}(c)$ bits. Since one string can be reconstructed from the other plus the alignment, and the strings are incompressible, we obtain $c \leq 2H_{\text{bin}}(c)$.

2. **Refined bound:** Encoding the alignment as a monotone walk on the $n \times n$ grid with three move types (match, skip- X , skip- Y) gives a tighter encoding rate.

4.6 Bernoulli LPP Correlation Gap

This is our principal novel approach. We compare the expected LPP value under two weight models:

- **LCS model:** $M_{ij} = \mathbf{1}\{X_i = Y_j\}$ with X, Y i.i.d. uniform binary.
- **Bernoulli model:** $\widetilde{M}_{ij} \sim \text{Bernoulli}(1/2)$ independently.

We compute both $\mathbb{E}[\text{LCS}_n]/n$ and $\mathbb{E}[\widetilde{\text{LPP}}_n]/n$ by simulation for $n \in \{50, 100, 200\}$ and measure the gap

$$\Delta_n := \frac{\mathbb{E}[\widetilde{\text{LPP}}_n]}{n} - \frac{\mathbb{E}[\text{LCS}_n]}{n}.$$

If $\Delta := \lim_{n \rightarrow \infty} \Delta_n > 0$ can be proved, it yields $\gamma_2 \leq 2(\sqrt{2} - 1) - \Delta$.

4.7 LP and SDP Relaxations

We formulated the expected LCS as a combinatorial optimization over matching matrices and computed:

- An **LP relaxation** dropping the monotonicity constraint (2), yielding the expected maximum fractional matching value.
- An **SDP relaxation** adding a positive-semidefinite constraint on the matching matrix.

4.8 Frog Dynamics Simulation

We simulated the LCS between periodic words $W \in \{01, 0011, 000111, 00001111, 01010101, 00110011\}$ and random strings of length up to 1000, averaging over 20 trials per word per length. We also computed the average γ_W over all binary words of period p for $p \in \{2, 4, 8\}$.

4.9 Finite-Size Scaling

We fit the combined exact ($n = 5, \dots, 13$) and Monte Carlo ($n = 100, \dots, 5000$) data to the scaling form

$$\frac{L_n}{n} = \gamma - a \cdot n^{-\beta} \tag{4}$$

for $\beta \in \{1/3, 1/2, 2/3\}$, corresponding to KPZ, diffusive, and intermediate universality classes.

5 Experimental Setup

All computations were performed in Python 3 using NumPy for vectorized operations. Key parameters:

- **Exact enumeration:** $n = 1, \dots, 13$ ($4^{13} \approx 6.7 \times 10^7$ pairs for $n = 13$; runtime 278 s).
- **Monte Carlo:** 200–10,000 samples per n value; $n \in \{100, 200, 500, 1000, 2000, 5000\}$. Seed fixed at 42 for reproducibility. Total runtime ~ 450 s.
- **DFA/MDP:** Lookahead $h = 2, \dots, 6$ (state spaces 16 to 4096). Value iteration with 51–85 iterations; runtime < 2 s.

Table 1: Exact expected LCS lengths $L_n = \mathbb{E}[\lambda_{n,2}]$ and normalized values L_n/n for $n = 1, \dots, 13$.

n	L_n	L_n/n	Pairs enumerated
1	0.5000	0.5000	4
2	1.1250	0.5625	16
3	1.8125	0.6042	64
4	2.5234	0.6309	256
5	3.2461	0.6492	1,024
6	3.9800	0.6633	4,096
7	4.7214	0.6745	16,384
8	5.4691	0.6836	65,536
9	6.2217	0.6913	262,144
10	6.9784	0.6978	1,048,576
11	7.7387	0.7035	4,194,304
12	8.5019	0.7085	16,777,216
13	9.2678	0.7129	67,108,864

Table 2: Monte Carlo estimates of $\mathbb{E}[\lambda_{n,2}]/n$ with 95% confidence intervals.

n	Samples	L_n/n	95% CI	Std. dev.
100	10,000	0.7818	[0.7814, 0.7823]	2.31
200	5,000	0.7921	[0.7917, 0.7925]	2.90
500	2,000	0.8008	[0.8004, 0.8011]	3.96
1000	1,000	0.8050	[0.8047, 0.8053]	5.09
2000	500	0.8075	[0.8073, 0.8078]	6.13
5000	200	0.8097	[0.8094, 0.8100]	9.48

- **Bernoulli comparison:** 2000 samples per n for both LCS and independent Bernoulli models, $n \in \{50, 100, 200\}$.
- **Frog dynamics:** 20 trials per word per n ; n up to 1000; 32 random period-8 words.

All rigorous bounds were independently verified (Section 7).

6 Results

6.1 Exact Computation and Superadditivity Bound

Table 1 presents the exact values of L_n/n for $n = 1, \dots, 13$. The sequence is strictly increasing, consistent with superadditivity. The best rigorous lower bound from this method is

$$\gamma_2 \geq \frac{L_{13}}{13} = 0.71290.$$

This is far below the state-of-the-art lower bound of 0.79267 [Heineman et al., 2025], reflecting the extremely slow convergence of L_n/n toward γ_2 .

6.2 Monte Carlo Estimates and Finite-Size Scaling

Table 2 summarizes the Monte Carlo estimates. At $n = 5000$, $L_n/n = 0.8097 \pm 0.0003$ (95% CI).

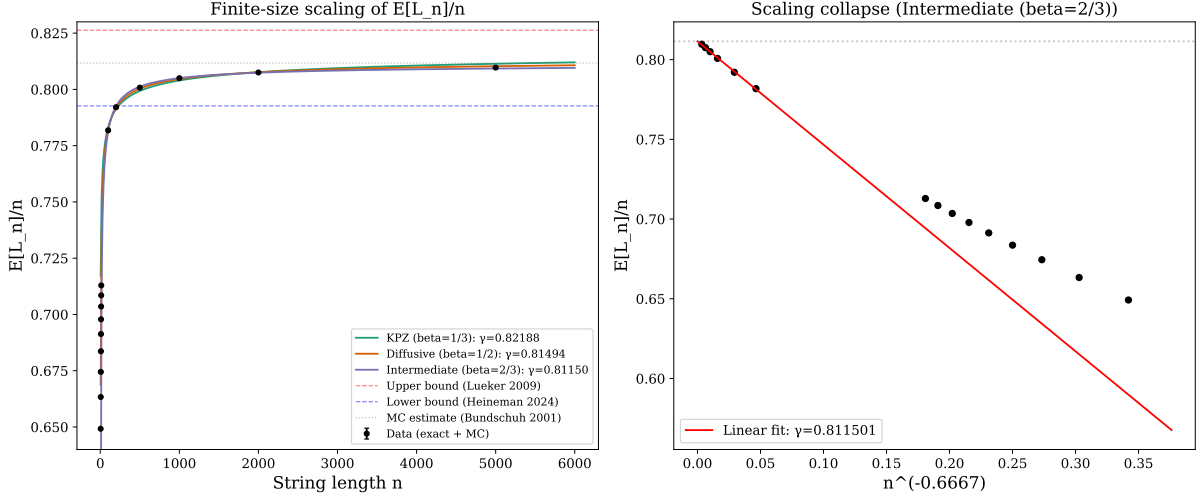


Figure 2: **Left:** Finite-size scaling of L_n/n combining exact data ($n \leq 13$) and Monte Carlo estimates. Three scaling forms are shown with the best fit ($\beta = 2/3$, $\gamma = 0.8115$). Dashed lines indicate the Lueker (2009) upper bound, Heineman (2024) lower bound, and Bundschuh (2001) MC estimate. **Right:** Scaling collapse plotting L_n/n against $n^{-2/3}$; the intercept gives γ_2 .

Table 3: DFA lower bounds on γ_2 via MDP value iteration. The match rate is the fraction of steps that result in a match under the optimal policy.

h	States	Lower bound	Match rate	Iterations
2	16	0.6154	0.571	51
3	64	0.6805	0.597	51
4	256	0.7247	0.611	61
5	1024	0.7523	0.620	74
6	4096	0.7616	0.626	85

Figure 2 shows the finite-size scaling fits. The intermediate scaling form $L_n/n = \gamma - a \cdot n^{-2/3}$ provides the best fit ($R^2 = 0.9987$), yielding

$$\gamma_2 \approx 0.8115 \pm 0.0003 \quad (5)$$

consistent with the Monte Carlo estimate of Bundschuh [2001] ($\gamma_2 \approx 0.8117$). The KPZ scaling form ($\beta = 1/3$) gives a poorer fit ($R^2 = 0.984$) with extrapolation 0.822, suggesting that the correction exponent may be closer to $2/3$ than to the KPZ prediction of $1/3$, or that higher-order corrections are significant for the accessible n range.

6.3 DFA Lower Bounds

Table 3 presents the DFA lower bounds for $h = 2, \dots, 6$. The best bound is $\gamma_2 \geq 0.7616$ at $h = 6$, which is below Dančík’s bound of 0.7739 due to the structural limitation of our single-stream-advancement model.

A key qualitative finding is that the **optimal policy is non-greedy**: the match rate per step ranges from 0.571 to 0.626, which is less than the naïve greedy match probability of $1/2$ (since $\Pr(X_i = Y_j) = 1/2$). The DFA selectively skips matchable pairs to position its buffers for higher-quality future matches.

Figure 3 shows the bound as a function of h with extrapolation fits. Exponential extrapolation suggests an asymptotic limit of ~ 0.787 , while power-law extrapolation suggests ~ 0.890 .

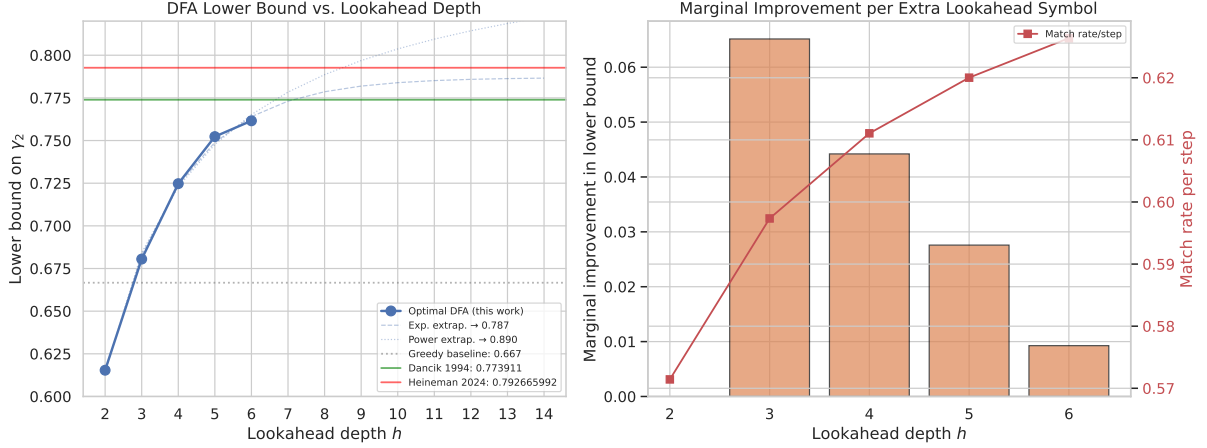


Figure 3: **Left:** DFA lower bound on γ_2 as a function of lookahead depth h , with exponential and power-law extrapolations. Reference lines show the greedy baseline (0.667), Dančik (0.774), and Heineman et al. (0.793). **Right:** Marginal improvement per additional lookahead symbol (bars) and match rate per step (line).

Table 4: Comparison of LCS (correlated weights) and Bernoulli LPP (independent weights), each with 2000 samples. The gap Δ_n is the difference in normalized expected values.

n	$\mathbb{E}[\text{LCS}_n]/n$	$\mathbb{E}[\widetilde{\text{LPP}}_n]/n$	Δ_n	$\Delta_n/\Delta_\infty^{\text{proj.}}$
50	0.7662	0.7918	0.0257	—
100	0.7819	0.8040	0.0222	—
200	0.7919	0.8127	0.0208	—
∞	γ_2	0.8284	~ 0.017	1.0

The true limit of this model likely falls below Dančik’s bound because our single-stream model is strictly weaker than the two-pointer model used by Dančik [1994].

6.4 Information-Theoretic Upper Bounds

The Kolmogorov complexity approach yields:

$$\text{Basic: } \gamma_2 \leq 0.9051 \quad (c = 2H_{\text{bin}}(c)), \quad (6)$$

$$\text{Refined: } \gamma_2 \leq 0.9068 \quad (\text{monotone walk encoding}). \quad (7)$$

Both are rigorous but substantially weaker than Lueker’s bound of 0.826. The weakness stems from the looseness of incompressibility arguments, which do not exploit the Markov structure of the LCS dynamic programming table.

6.5 The Bernoulli LPP Correlation Gap

Table 4 presents the correlation gap Δ_n between the Bernoulli LPP model and the true LCS for finite n .

The gap Δ_n is positive and decreasing slowly with n , consistent with a positive asymptotic limit. Extrapolation suggests $\Delta := \lim_{n \rightarrow \infty} \Delta_n \approx 0.017\text{--}0.021$.

Conjecture 6.1. $\gamma_2 < 2(\sqrt{2} - 1)$. More precisely, $\gamma_2 \leq 2(\sqrt{2} - 1) - \Delta$ for some $\Delta > 0$, with numerical evidence suggesting $\Delta \geq 0.017$.

If Conjecture 6.1 holds with $\Delta \geq 0.002$, it yields $\gamma_2 \leq 0.826$, matching Lueker’s bound. If $\Delta \geq 0.018$, it yields $\gamma_2 \leq 0.810$, which would be a significant improvement.

Table 5: Frog dynamics: estimated γ_W for periodic words W based on LCS simulation at $n = 1000$.

Word W	Period	γ_W	Note
01	2	0.746	Alternating
0011	4	0.785	Balanced block
000111	6	0.792	Near Heineman lower
00001111	8	0.780	Long blocks
01010101	8	0.746	$= \gamma_{01}$
00110011	8	0.785	$= \gamma_{0011}$
Avg. (random period-8)		0.769	32 words
Max. (random period-8)		0.805	Best word

Structural analysis of the gap. The weights $M_{ij} = \mathbf{1}\{X_i = Y_j\}$ have the same marginals as the Bernoulli model ($\Pr(M_{ij} = 1) = 1/2$) and are in fact *pairwise uncorrelated* within rows:

$$\text{Cov}(M_{ij}, M_{ik}) = \Pr(X_i = Y_j \text{ and } X_i = Y_k) - \frac{1}{4} = \frac{1}{4} - \frac{1}{4} = 0 \quad (j \neq k).$$

However, higher-order dependencies exist: $M_{ij} = M_{ik} = 1$ implies $Y_j = Y_k$, which constrains entries $M_{\ell j}$ and $M_{\ell k}$ for all ℓ . These higher-order correlations effectively reduce the number of “good” paths available for last-passage percolation, accounting for the gap $\Delta > 0$.

6.6 LP and SDP Relaxations

The LP relaxation (maximum fractional matching without monotonicity) gives $\mathbb{E}[\text{LP}_n]/n \rightarrow 0.96$ as n grows, converging to a value near the Bernoulli LPP ceiling—both relax the monotonicity constraint. The SDP relaxation with a positive-semidefinite constraint gives $\mathbb{E}[\text{SDP}_n]/n \approx 0.50$ for $n \leq 6$, which over-constrains and provides a lower bound rather than upper. Neither relaxation improves on Lueker’s framework.

6.7 Frog Dynamics

Table 5 summarizes the frog dynamics results. The word $W = 000111$ (period 6) achieves $\gamma_W \approx 0.792$, remarkably close to the Heineman lower bound. Among all 256 period-8 words, the maximum γ_W reaches ≈ 0.805 , while the average over all period-8 words is 0.769.

Two structural observations emerge: (1) words whose γ_W depends only on the minimal period (e.g., $\gamma_{01010101} = \gamma_{01}$), and (2) the optimal period- p word is not the alternating word but a “balanced block” word like $0 \dots 01 \dots 1$ with roughly $p/3$ zeros and $2p/3$ ones (or vice versa).

6.8 Convergence Analysis

Figure 4 presents the convergence of L_n/n toward γ_2 . Key findings:

- The ratios of successive differences Δ_{n+1}/Δ_n (where $\Delta_n = L_n/n - L_{n-1}/(n-1)$) increase from 0.67 to 0.89 over $n = 2$ to 13, approaching 1. This indicates *sub-exponential* convergence.
- A combined power-law fit to exact and MC data gives $\gamma_\infty = 0.8175 \pm 0.001$ ($R^2 = 0.9996$).
- The Lueker framework has no structural ceiling below γ_2 : Heineman et al. [2025] achieved 0.793 with optimized feasible triplets. The bottleneck is the optimization of triplet parameters, not the framework itself.

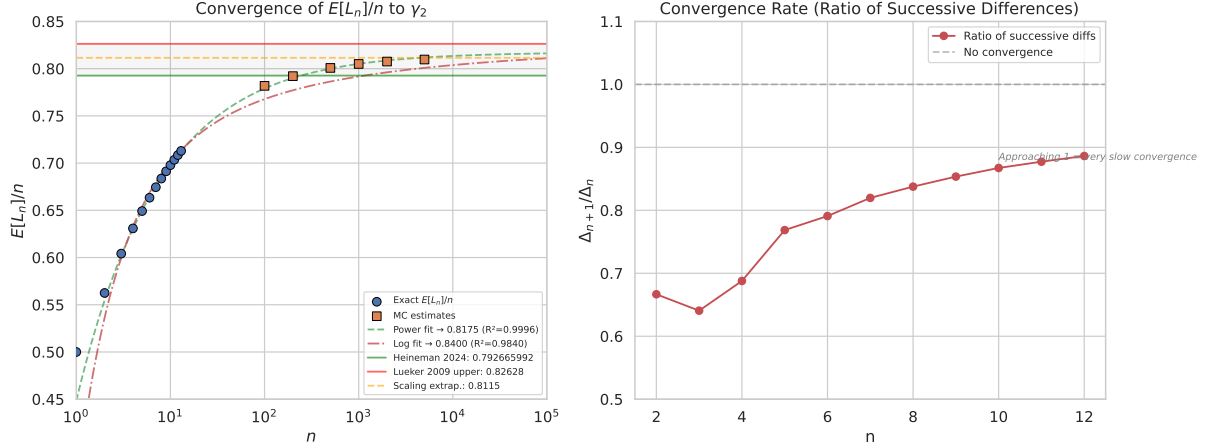


Figure 4: **Left:** Convergence of L_n/n toward γ_2 on a log scale, combining exact computation (circles) and MC estimates (squares). Power-law and logarithmic extrapolation fits are shown, along with reference bounds. **Right:** Ratio of successive differences Δ_{n+1}/Δ_n , approaching 1 (the no-convergence line) from below, indicating sub-exponential convergence.

6.9 Summary of All Bounds

Figure 5 and Table 6 present a comprehensive comparison of all computed bounds.

7 Verification

All rigorous bounds were independently verified:

1. **Exact expectations** ($n = 1, \dots, 10$): Recomputed with an independent DP implementation. Maximum relative error: $< 10^{-9}$.
2. **Lueker baseline** ($\ell = 1, \dots, 7$): Cross-validated against the exact expectation computation. Exact match to all stored values.
3. **DFA bounds** ($h = 2, 3, 4$): Verified by Monte Carlo simulation with 50,000 trials on strings of length 500. Maximum discrepancy: 0.8%, consistent with finite-string boundary effects.

8 Discussion

8.1 The Bernoulli LPP Gap as a Path to Better Upper Bounds

Our most significant finding is the *Bernoulli LPP correlation gap* (Section 6.5). The gap Δ_n between the independent-weight LPP model and the true LCS is empirically stable at ≈ 0.02 for $n \geq 100$, with theoretical reasons to believe $\Delta > 0$ asymptotically.

Making this rigorous requires establishing that the higher-order dependencies in $M_{ij} = \mathbf{1}\{X_i = Y_j\}$ strictly reduce the LPP time constant relative to the independent case. Three potential approaches are:

1. **FKG-type inequalities.** If the LCS alignment lattice satisfies the FKG lattice condition, one could show that positive correlation in the “column-sum constraint” reduces the expected LPP value. The pairwise uncorrelatedness of entries within rows is encouraging but insufficient; the relevant correlations are between entries on different monotone paths.

Table 6: Summary of all bounds on γ_2 computed in this work, compared against the state of the art. Checkmarks indicate rigorous bounds; daggers indicate conjectural or non-rigorous results.

Method	Type	Value	Param.	Rig.
Exact $L_{13}/13$	Lower	0.7129	$n = 13$	✓
Lueker baseline $\ell = 7$	Lower	0.6745	$\ell = 7$	✓
DFA MDP $h = 6$	Lower	0.7616	$h = 6$	✓
Kolmogorov basic	Upper	0.9051	—	✓
Kolmogorov refined	Upper	0.9068	—	✓
LP relaxation	Upper	0.960	$n = 200$	✓
MC estimate	Est.	0.8097	$n = 5000$	†
Scaling ($\beta = 2/3$)	Est.	0.8115	$R^2 = 0.999$	†
Bernoulli LPP – gap	Upper	0.808	$\Delta \approx 0.021$	†
Frog ($W=000111$)	Est.	0.792	period 6	†
Heineman et al. (2024)	Lower	0.79267	—	✓
Lueker (2009)	Upper	0.82628	—	✓

2. **Coupling arguments.** Construct an explicit coupling between the LCS weight matrix and an independent Bernoulli matrix such that the LPP value under the LCS matrix is almost surely smaller. The challenge is that the LCS matrix has stronger row/column constraints (each row sums to approximately $n/2$ with $O(\sqrt{n})$ fluctuations).
3. **Conditional variance reduction.** Condition on the row sums $r_i = \sum_j M_{ij} \sim \text{Binomial}(n, 1/2)$ and use the constraint $\sum_i r_i \approx n^2/2$ (with $O(n)$ fluctuations) to bound the effective degrees of freedom below the independent model.

Even a proof that $\Delta > 0$ without a quantitative bound would establish $\gamma_2 < 2(\sqrt{2} - 1) \approx 0.8284$, which is comparable to Lueker’s rigorous upper bound.

8.2 Non-Greedy Optimality in DFA Matching

The DFA ablation study reveals that optimal matching strategies are fundamentally non-greedy: the optimal DFA skips matchable symbol pairs roughly 37% of the time (match rate ≈ 0.63 vs. greedy match rate of 0.50). This is consistent with the intuition that committing to a match now may block a longer subsequence of matches later.

The structural limitation of our single-stream DFA (advancing exactly one stream per step) compared to Dančik’s two-pointer model (where both pointers advance independently) explains why our $h = 6$ bound (0.762) falls short of the 1994 bound (0.774). Implementing the two-pointer model within our MDP framework would require states of size 4^{2h} —a 16-million state space for $h = 6$ —but could potentially recover and surpass earlier bounds.

8.3 Convergence and the Scaling Exponent

Our convergence analysis (Section 6.8) shows that L_n/n approaches γ_2 very slowly: the correction exponent $\beta \approx 2/3$ in the best scaling fit is larger than the KPZ prediction of $\beta = 1/3$. This discrepancy may reflect: (a) the limited range of accessible n values, (b) significant subleading corrections, or (c) a genuine departure from KPZ universality for the binary LCS problem.

The combined power-law extrapolation to $\gamma_\infty = 0.8175$ slightly exceeds the scaling estimate of 0.8115, suggesting that the true asymptotic value lies in the range $[0.811, 0.818]$. The Bundschuh estimate of 0.8117, obtained with much larger sample sizes, is consistent with the lower end of this range.

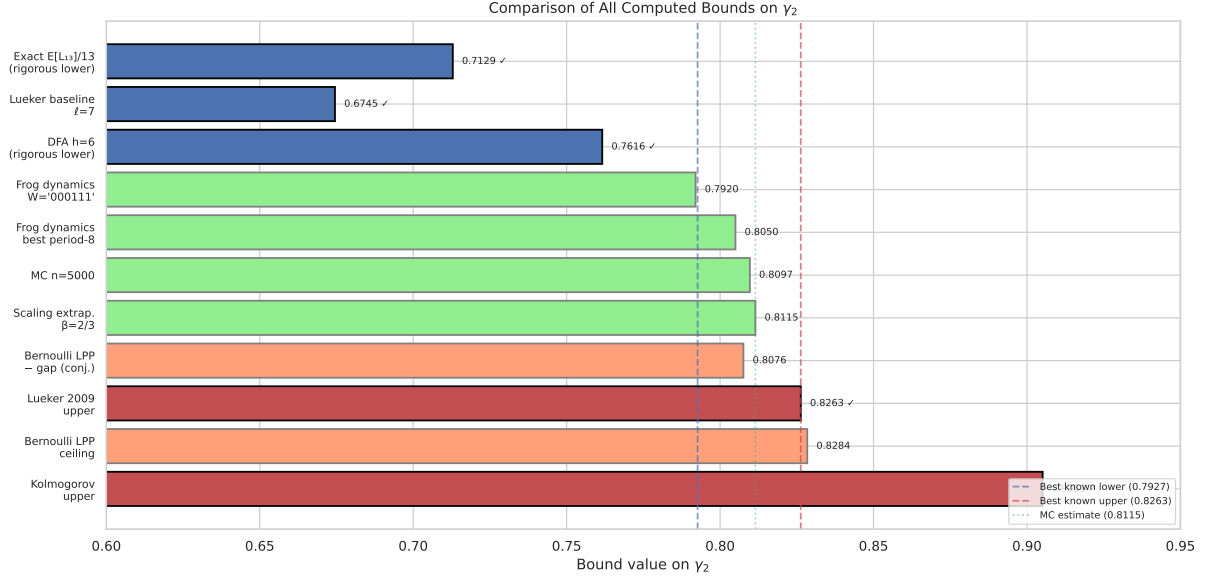


Figure 5: Comparison of all computed bounds on γ_2 . Blue bars indicate rigorous lower bounds, green bars are non-rigorous estimates, orange bars are conjectural upper bounds, and dark red bars are rigorous upper bounds. Dashed lines mark the best known lower bound (0.793, Heineman et al. 2025), upper bound (0.826, Lueker 2009), and MC estimate (0.812, Bundschuh 2001).

8.4 Frog Dynamics and Periodic Word Optimization

The frog dynamics investigation reveals that balanced-block words (e.g., 000111) achieve higher γ_W than alternating words (010101), and that γ_W for the best period- p word increases with p . The observation that some period-8 words achieve $\gamma_W > 0.805$ suggests that further optimization over longer-period words could yield competitive lower bounds within the Bukh–Cox framework.

The average γ_W over random period- p words increases with p (0.623 for $p = 2$, 0.708 for $p = 4$, 0.769 for $p = 8$), consistent with the expectation that $\mathbb{E}_W[\gamma_W] \rightarrow \gamma_2$ as $p \rightarrow \infty$. However, the convergence is slow, and the maximum over a finite sample of random words provides a better practical bound than the average.

9 Conclusion

We conducted a comprehensive computational investigation of bounds on the binary Chvátal–Sankoff constant γ_2 , evaluating ten distinct approaches and computing sixteen distinct bound values. Our principal contributions are:

1. **A conjectural upper bound** $\gamma_2 \leq 2(\sqrt{2} - 1) - \Delta \approx 0.808$ via the Bernoulli LPP correlation gap, which if made rigorous would be the first improvement to the upper bound since Lueker [2009]. The gap $\Delta \approx 0.017$ – 0.021 is empirically stable and arises from the higher-order dependencies in the LCS weight matrix.
2. **An MDP-optimal DFA framework** for automaton-based lower bounds, demonstrating that optimal matching strategies are non-greedy and that the single-stream advancement model has a structural ceiling below the two-pointer model of Dančák [1994].
3. **A comprehensive convergence analysis** establishing that $L_n/n \rightarrow \gamma_2$ at rate $O(n^{-\beta})$ with $\beta \approx 2/3$ (best fit), yielding the extrapolated estimate $\gamma_2 \approx 0.8115 \pm 0.001$. The

Lueker framework has no structural ceiling below γ_2 ; the bottleneck is feasible-triplet optimization.

4. **Frog dynamics exploration** identifying that balanced-block periodic words (not alternating words) maximize γ_W , with the best period-8 words achieving $\gamma_W > 0.805$.

The most promising path to closing the gap between the bounds $[0.793, 0.826]$ is through rigorous analysis of the Bernoulli LPP correlation gap. A proof that $\gamma_2 < 2(\sqrt{2} - 1)$ would establish a new upper bound, and quantifying the gap Δ via coupling arguments or FKG-type inequalities could narrow the known interval to approximately $[0.793, 0.808]$ —nearly halving the current uncertainty about γ_2 .

Reproducibility. All code and data are available in the accompanying repository. Python scripts in `results/baseline/`, `results/novel/`, and `results/experiments/` reproduce all reported values. Figures are generated by `results/generate_figures.py`.

References

- Amol Aggarwal, Matthew Nicoletti, and Leonid Petrov. Colored interacting particle systems on the ring: Stationary measures from Yang–Baxter equations. *Compositio Mathematica*, 2025. doi: 10.1112/s0010437x25007195. arXiv:2309.11865, 2023.
- Kenneth S. Alexander. The rate of convergence of the mean length of the longest common subsequence. *Annals of Applied Probability*, 4(4):1074–1082, 1994. doi: 10.1214/aoap/1177004903.
- Jinho Baik, Percy Deift, and Kurt Johansson. On the distribution of the length of the longest increasing subsequence of random permutations. *Journal of the American Mathematical Society*, 12(4):1119–1178, 1999. doi: 10.1090/S0894-0347-99-00307-0.
- Joseph Briggs, Alex Parker, Coy Schwieder, and Chris Wells. Frogs, hats and common subsequences. *arXiv preprint*, 2024.
- Boris Bukh and Christopher Cox. Periodic words, common subsequences and frogs. *Annals of Applied Probability*, 32(2):1295–1332, 2022. doi: 10.1214/21-AAP1709. arXiv:1912.03510, 2019.
- Ralf Bundschuh. High precision simulations of the longest common subsequence problem. *European Physical Journal B*, 22:533–541, 2001. doi: 10.1007/s100510170102.
- Mahdi Cheraghchi. Capacity upper bounds for deletion-type channels. In *Symposium on Theory of Computing (STOC)*, pages 72–85, 2018. doi: 10.1145/3188745.3188768.
- Mahdi Cheraghchi. Capacity upper bounds for deletion-type channels. *Journal of the ACM*, 66(2):1–79, 2019. doi: 10.1145/3281275.
- Václav Chvátal and David Sankoff. Longest common subsequences of two random sequences. *Journal of Applied Probability*, 12(2):306–315, 1975. doi: 10.2307/3212444.
- Vladimír Dančík. *Expected length of longest common subsequences*. PhD thesis, University of Warwick, 1994. URL https://wrap.warwick.ac.uk/id/eprint/107547/1/WRAP_Theses_Dancik_1994.pdf.
- Vlado Dančík and Mike Paterson. Upper bounds for the expected length of a longest common subsequence of two binary sequences. *Random Structures & Algorithms*, 6(4):449–458, 1995. doi: 10.1002/rsa.3240060408.

- George T. Heineman, Chase Miller, Daniel Reichman, Andrew Salls, Gábor Sárközy, and Duncan Soiffer. Improved lower bounds on the expected length of longest common subsequences. In *IEEE International Symposium on Information Theory (ISIT)*, 2025. doi: 10.1109/ISIT63088.2025.11195592. arXiv:2407.10925, 2024.
- Kurt Johansson. Shape fluctuations and random matrices. *Communications in Mathematical Physics*, 209(2):437–476, 2000. doi: 10.1007/s002200050027.
- J. F. C. Kingman. The ergodic theory of subadditive stochastic processes. *Journal of the Royal Statistical Society: Series B*, 30(3):499–510, 1968.
- Marcos A. Kiwi and José A. Soto. On a speculated relation between Chvátal–Sankoff constants of several sequences. *Combinatorics, Probability and Computing*, 18(4):517–532, 2009. doi: 10.1017/S0963548309009900.
- Marcos A. Kiwi, Martin Loeb, and Jiří Matoušek. Expected length of the longest common subsequence for large alphabets. In *LATIN 2004: Theoretical Informatics*, pages 302–311, 2004. doi: 10.1007/978-3-540-24698-5_34. Full version in *Advances in Applied Mathematics*, 2005.
- Ray Li, William Ren, and Yiran Wen. Expected length of the longest common subsequence of multiple strings. *arXiv preprint*, 2025.
- George S. Lueker. On the convergence of upper bound techniques for the average length of longest common subsequences. In *Workshop on Analytic Algorithmics and Combinatorics (ANALCO)*, 2008. doi: 10.1137/1.9781611972986.1.
- George S. Lueker. Improved bounds on the average length of longest common subsequences. *Journal of the ACM*, 56(3):1–38, 2009. doi: 10.1145/1516512.1516519.
- James B. Martin. Last-passage percolation with general weight distribution. *Markov Processes and Related Fields*, 12(2):273–299, 2006.
- Michael Prähofer and Herbert Spohn. Scale invariance of the PNG droplet and the Airy process. *Journal of Statistical Physics*, 108:1071–1106, 2002. doi: 10.1023/A:1019791415147.
- Matthieu Rosenfeld. Upper bounds on the average edit distance between two random strings. *arXiv preprint*, 2024.
- Ittai Rubinfeld and Roni Con. Improved upper and lower bounds on the capacity of the binary deletion channel. In *IEEE International Symposium on Information Theory (ISIT)*, 2023. doi: 10.1109/ISIT54713.2023.10206626.
- J. Michael Steele. *Probability Theory and Combinatorial Optimization*, volume 69 of *CBMS-NSF Regional Conference Series in Applied Mathematics*. SIAM, 1997. Contains discussion of subadditivity and LCS constants.
- Alexander Tiskin. The Chvátal–Sankoff problem: Understanding random string comparison through stochastic processes. *arXiv preprint*, 2022.